Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Pure 1 – Taylor Series

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State Taylor's theorem for expansion about $x = a$. **[3]**

2. Find the Taylor series for e^x about $x = 1$ up to $(x-1)^3$. **[4]**

3. Find the Taylor series for $\sin x$ about $x = \frac{\pi}{2}$ up to the term in $\left(x - \frac{\pi}{2}\right)^3$. **[4]**

4. Find the Taylor series for $\ln x$ about $x = 1$ up to $(x-1)^3$. **[4]**

5. Expand $\cos x$ about $x = \frac{\pi}{4}$ up to the second degree term. **[4]**

6. Find the first three non-zero terms of the Taylor series for $\sqrt{x}$ about $x = 4$. **[4]**

7. Use Taylor series to approximate $\sin 1$ given expansion about 0. **[3]**

8. Find the Taylor polynomial of degree 2 for $f(x) = \frac{1}{x}$ about $x = 1$. **[3]**

9. Explain the difference between a Maclaurin series and a Taylor series. **[2]**

10. Find the Taylor series for e^x about $a = 2$ up to $(x-2)^2$. **[3]**

11. Expand $\tan x$ about $x = 0$ up to x^3 (Maclaurin). **[4]**

12. Find $f(2)$, $f'(2)$, $f''(2)$ if $f(x) = x^3 - 3x + 1$ and write the Taylor polynomial of degree 2 about 2. **[4]**

13. Use the Taylor expansion of $\ln(1 + x)$ about 0 to approximate $\ln 1.1$. **[3]**

14. Find the remainder term (Lagrange form) for the Taylor expansion of e^x after n terms. **[3]**

15. State when a Taylor series converges to the function. **[2]**

Section B – Hard (Q16–25)

16. Find the Taylor series for $\frac{1}{1+x}$ about $x = 1$ up to $(x - 1)^3$. **[5]**

17. Expand $f(x) = \sin x \cos x$ about $x = 0$ up to x^5. **[5]**

18. Find the Taylor series for $\sqrt{1 + x}$ about $x = 0$ up to x^3 and use it about a shifted point. **[5]**

19. Use Taylor expansion of e^{-x} about 0 to evaluate $\lim_{x\to 0} \frac{1-e^{-x}}{x}$. **[4]**

20. Find the cubic Taylor polynomial for $f(x) = \arctan x$ about $x = 0$. **[4]**

21. Expand $\ln(\cos x)$ about $x = 0$ up to x^4. **[6]**

22. Find the Taylor series for $f(x) = \frac{1}{\sqrt{x}}$ about $x = 1$ up to $(x - 1)^2$. **[4]**

23. Use Taylor's theorem to prove a small-angle approximation $\sin x \approx x - \frac{x^3}{6}$. **[3]**

24. Find the interval of convergence of the Taylor series for $\ln(1+x)$ about 0. **[3]**

25. Expand $f(x) = e^x \sin x$ about $x = 0$ up to x^4. **[6]**

End of Worksheet • *Show all working clearly.*